# Asymptotic analysis

## Programming · University · Lecture 1



# In this lecture

- Use O, Θ and Ω precisely.
- Solve recurrences.
- Teach growth by measuring.

# Remember

- Growth, not seconds.
- O is an upper bound.
- Measure, then define.

# Time it, then prove it

1. Warm-up (5 min): Guess the time
2. Measure (20 min): Two sorts.
3. Define (15 min): The notation.
4. Prove (15 min): The recurrence.
5. Wrap-up (5 min): One-minute paper.

# Let’s discuss

1. Where did the curves cross?
2. Why do constants vanish?
3. Which misconception did you hear?

# Exit ticket

- Write one true and one false statement using Big-O.